

SIDERIUS:

*Resource-Aware Multi-Agent System
for Automated Machine Learning in Science*

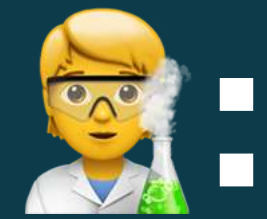
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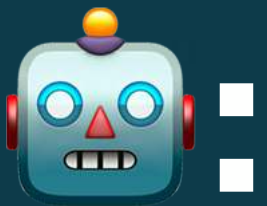
20260902@Fast Machine Learning for Science Conference 2026

Introduction: Background of Science Agents

The Ultimate Goal, a Big Dream



"Find dark matter for me"



Read papers

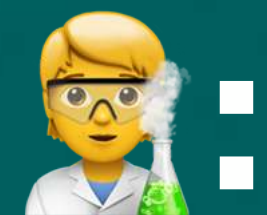
Run simulations

Optimize hardware

Result

Analyze data

Build experiment

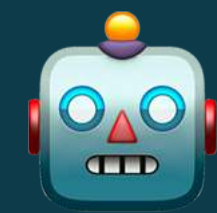


Good. Only ping me when you really find it

**A Few
~~Moments~~ Years Decades
Later...**

Introduction: Background of Science Agents

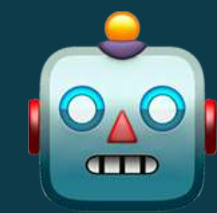
The Ultimate Goal, a Big Dream



: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

Introduction: Background of Science Agents

The Ultimate Goal, a Big Dream

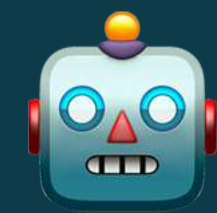


: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

Two Key Words in this Dream:

Introduction: Background of Science Agents

The Ultimate Goal, a Big Dream



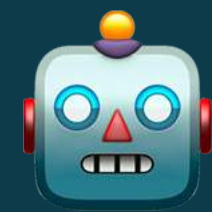
⚡: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

Two Key Words in this Dream:

End-to-end: Agents handles every stage, automates the loop

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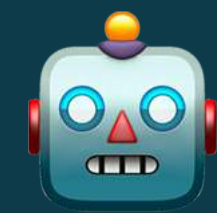
Two Key Words in this Dream:

End-to-end: Agents handles every stage, automates the loop

Long-horizon: Agents keep working for days, months, years

Introduction: Background of Science Agents

Yes, but...



: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

Introduction: Background of Science Agents

Yes, but...

🤖: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

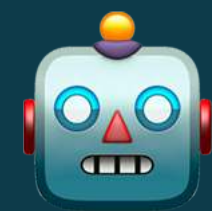
Case 1

🤖: Look at this awesome Axion that I found!
According to literature,
It is one of the dark matter candidates



Introduction: Background of Science Agents

Yes, but...



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Problem with this Dream:

Drifting/Gaming: When the goal is not clear (and it is always hard to have a 100% guarantee!), the agent optimizes endlessly, but chasing a phantom

Introduction: Background of Science Agents

Yes, but...

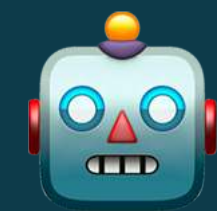
: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

Case 2

: Now the main problem is solved!
One last minor thing that I need you to do:
pay the 1 trillion bill for token usage, GPU computation, and instrumentation.

Introduction: Background of Science Agents

Yes, but...



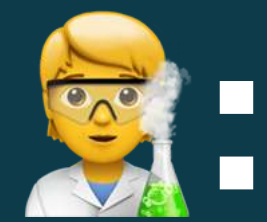
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Problem with this Dream:

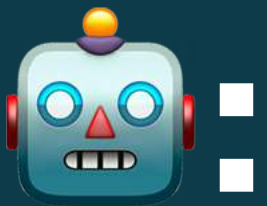
Wasting: Coming up with a huge amount of hypotheses is easy for LLM, verifying them is not. **Science is not simply brute-force repeating.**

Zoom-in: Automatic ML in Science

An Easier Task (Not Finding Dark Matter)



"Beat SOTA on this dataset, then write a paper for me"



Read papers

Design algorithm

Implement code

Paper

Analyze results

Train models

This is already happening.

The easiest case:

send **"/goal Beat SOTA on this dataset, then write a paper for me"** to codex

Zoom-in: Automatic ML in Science

An Easier Task (Not Finding Dark Matter)

Drifting/Gaming: Overfit, Mode Collapse, etc

Zoom-in: Automatic ML in Science

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Human-in-the-loop: adjust the road map; set guardrails

Zoom-in: Automatic ML in Science

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Drifting/Gaming: Overfit, Mode Collapse, etc

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Wasting: GPU is expensive, data storage is expensive too

Zoom-in: Automatic ML in Science

An Easier Task (Not Finding Dark Matter)

Drifting/Gaming: Overfit, Mode Collapse, etc

Human-in-the-loop: adjust the road map; set guardrails

Wasting: GPU is expensive, data storage is expensive too

Resource-awareness:

- Predict before execution
- Plan with the budget
- Reflect on the cost

Zoom-in: Automatic ML in Science

The current status, and gaps

	Domain workflow	Human-in-the-loop	Resource-aware
General agents Codex, Claude Code	○	●	◐
Specific agents Fixed workflow	●	◐	◐

Can we build something in between, and better?

The motivation of SIDERIUS

- Human-in-the-loop

Trade-off between flexibility and restriction

- Resource-awareness

LLM does a very bad guessing, unless we teach it explicitly

The Background of SIDERIUS

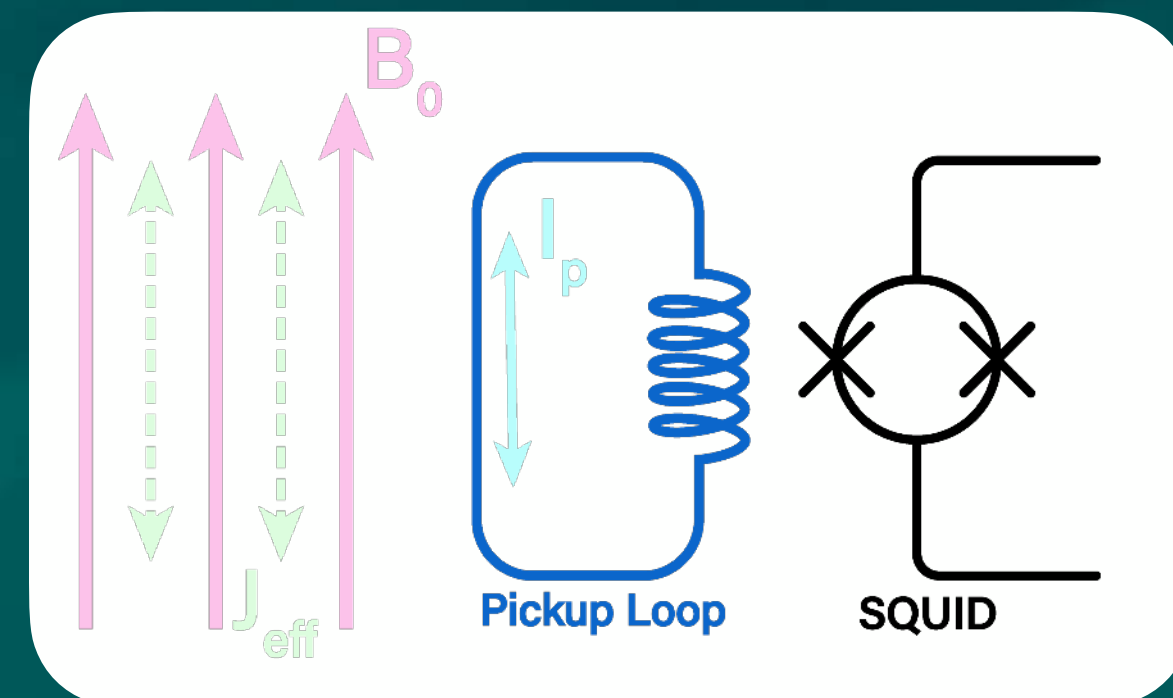
TIDMAD: A Sandbox for LLM Agent to Search for Dark Matter

Axion-Modified Maxwell's Equation
at $L \ll \lambda_a$:

$$\nabla \times \mathbf{B} \approx \mathbf{J}_{eff}$$
$$\mathbf{J}_{eff} = g_{a\gamma\gamma} \sqrt{2\rho_{DM}} \cos(m_a t) \mathbf{B}$$

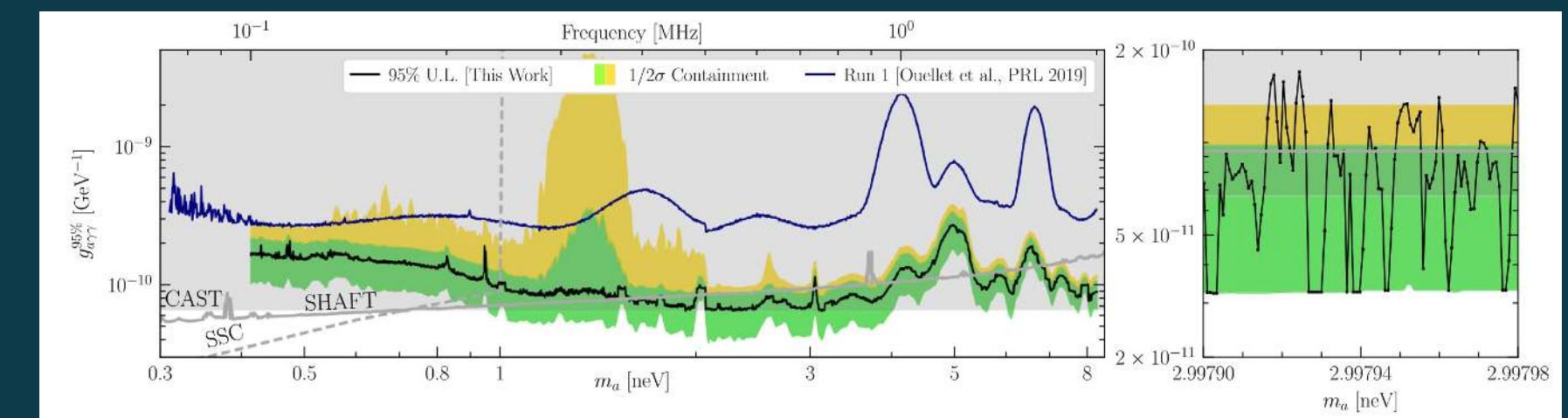


Magnetic Flux created
by the effective current



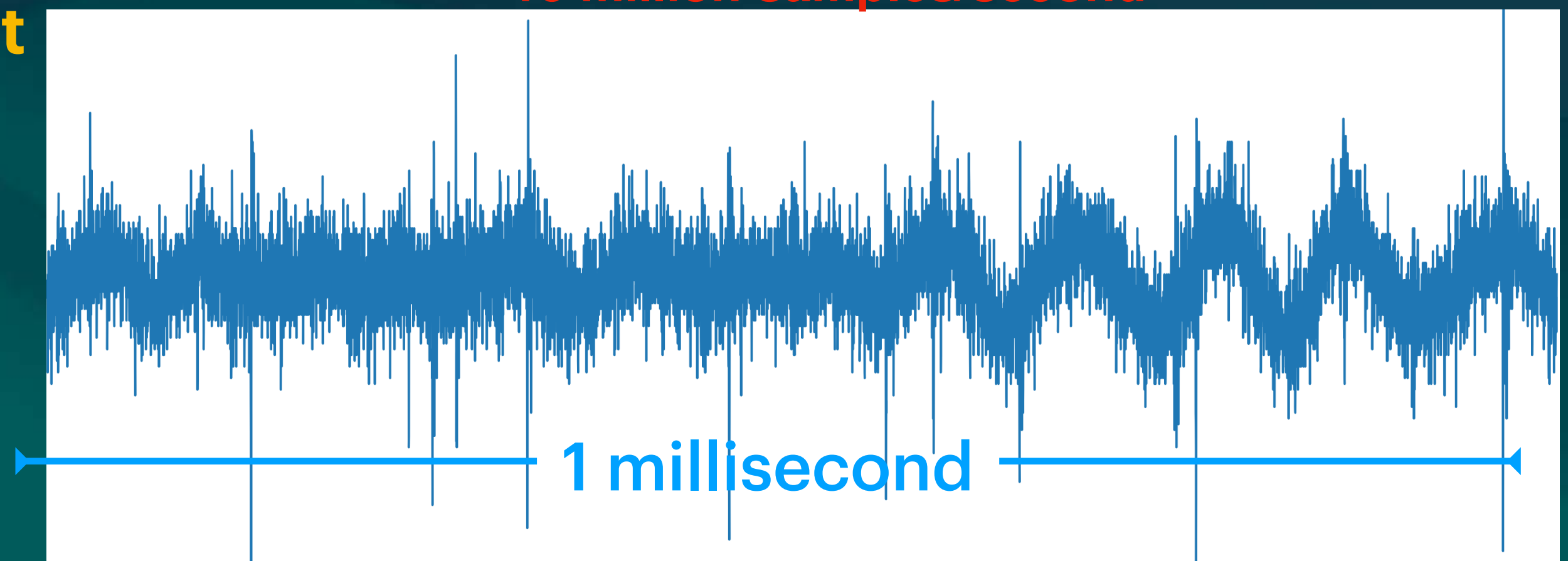
Frequency Spectrum

Broadband search for axion DM

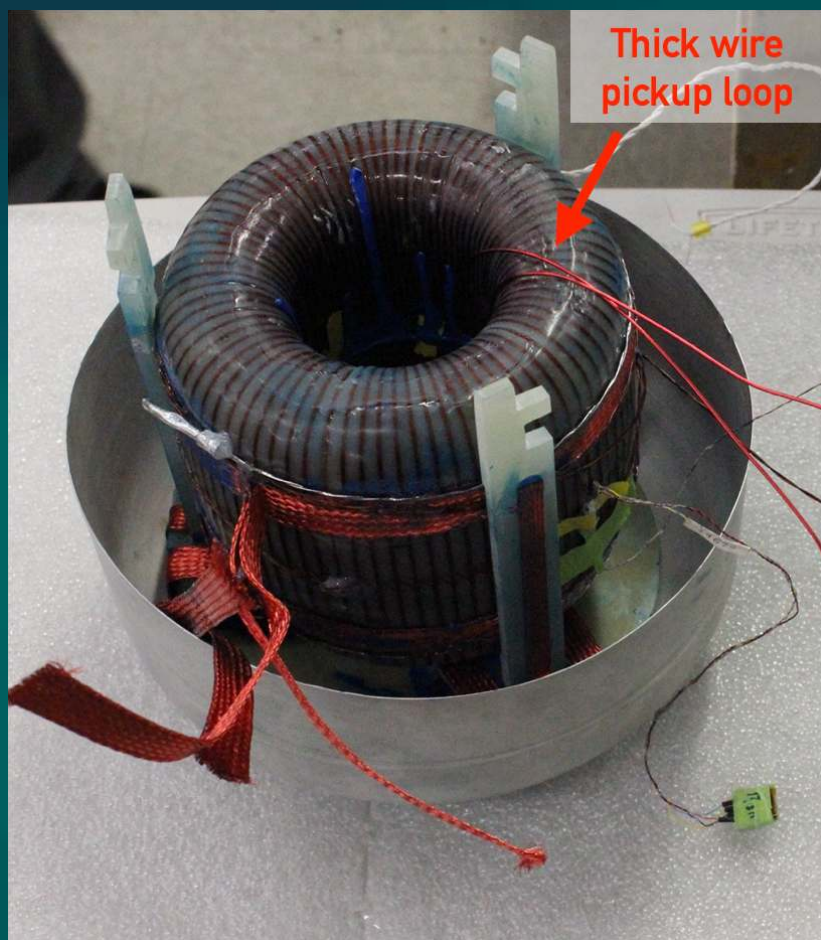


Ultra-long Time Series

10 million samples/second



Experimental Apparatus Constructed by Winslow Lab at MIT

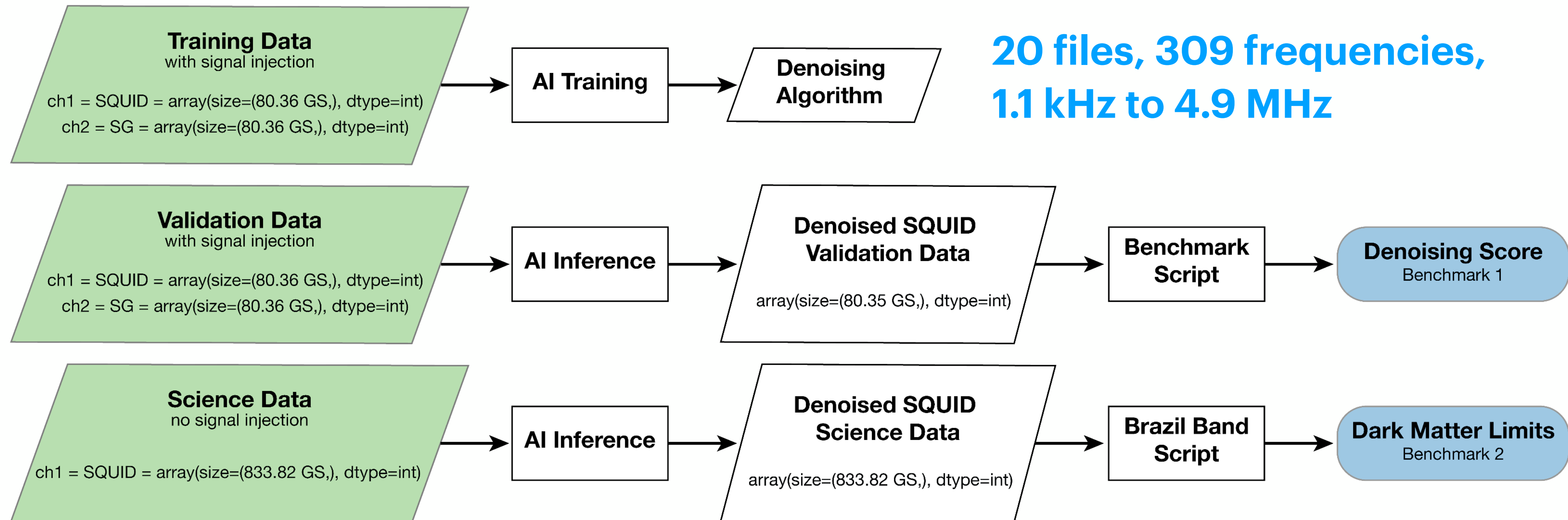


The Background of SIDERIUS

TIDMAD: A Sandbox for LLM Agent to Search for Dark Matter

Labeled Pair: [Signal only] — [Signal + noise]

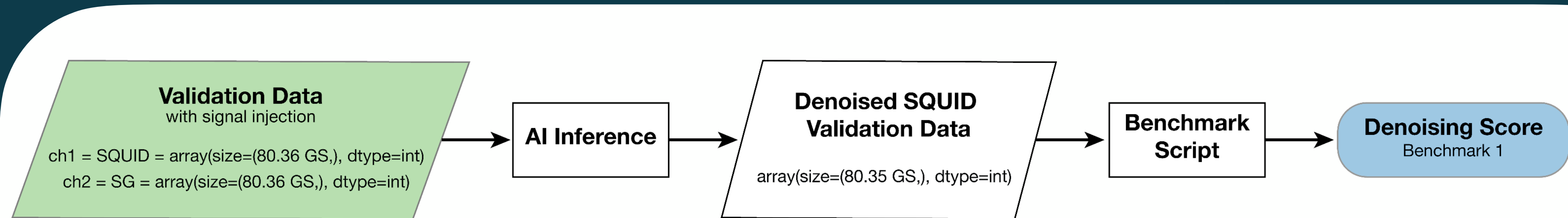
**20 files, 309 frequencies,
1.1 kHz to 4.9 MHz**



Search for dark matter in AI-denoised data

The Background of SIDERIUS

The Metric to Optimize in TIDMAD



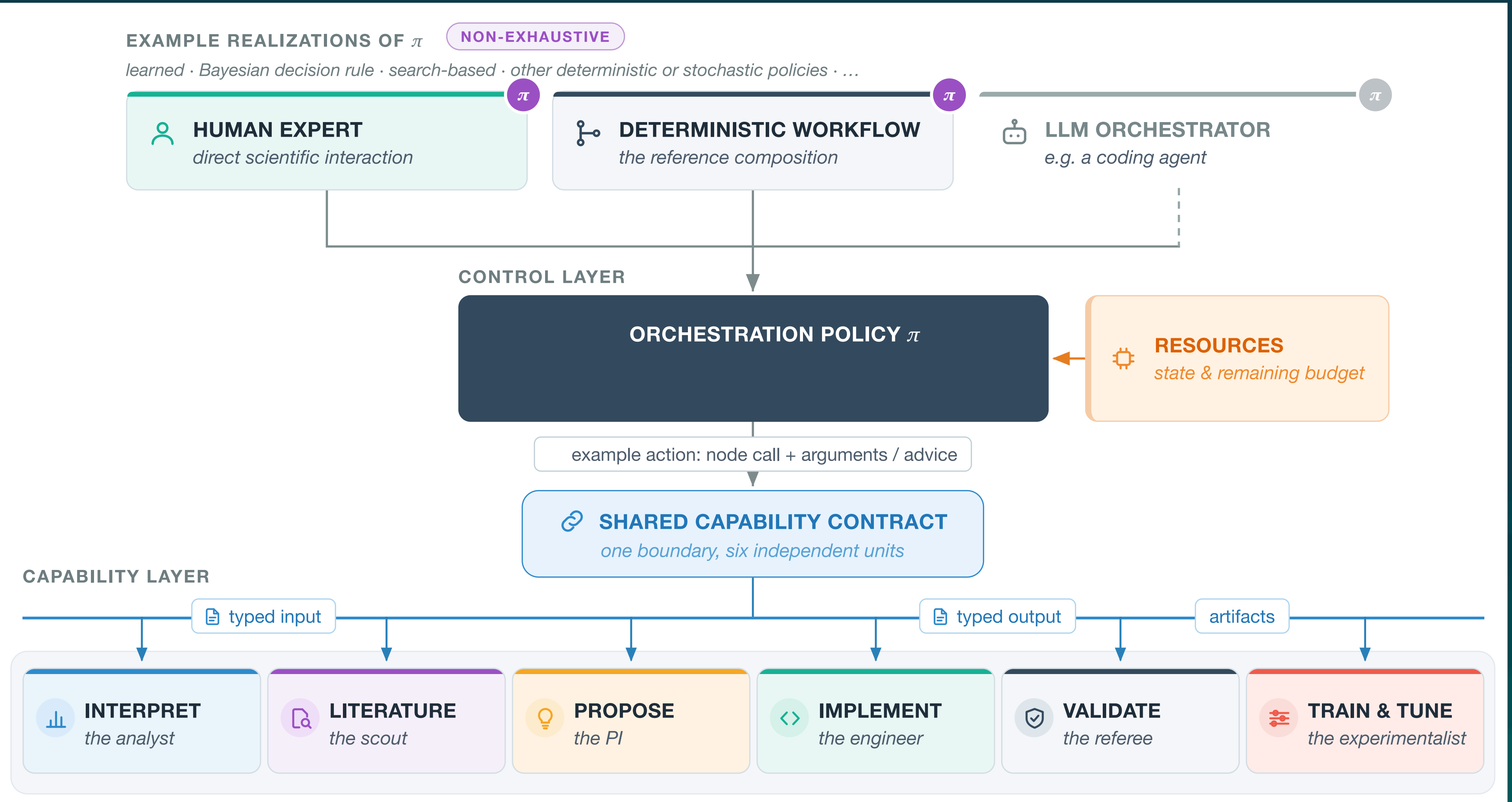
Motivation: Axion signal is a monochromatic wave

Denoising Score: Signal-to-noise Ratio in the frequency spectrum

$$\Lambda = \left(\frac{1}{n} \sum_{i=0}^n (SNR_{\text{SQUID}})_i \times (SNR'_{\text{Injected}})_i \right)$$

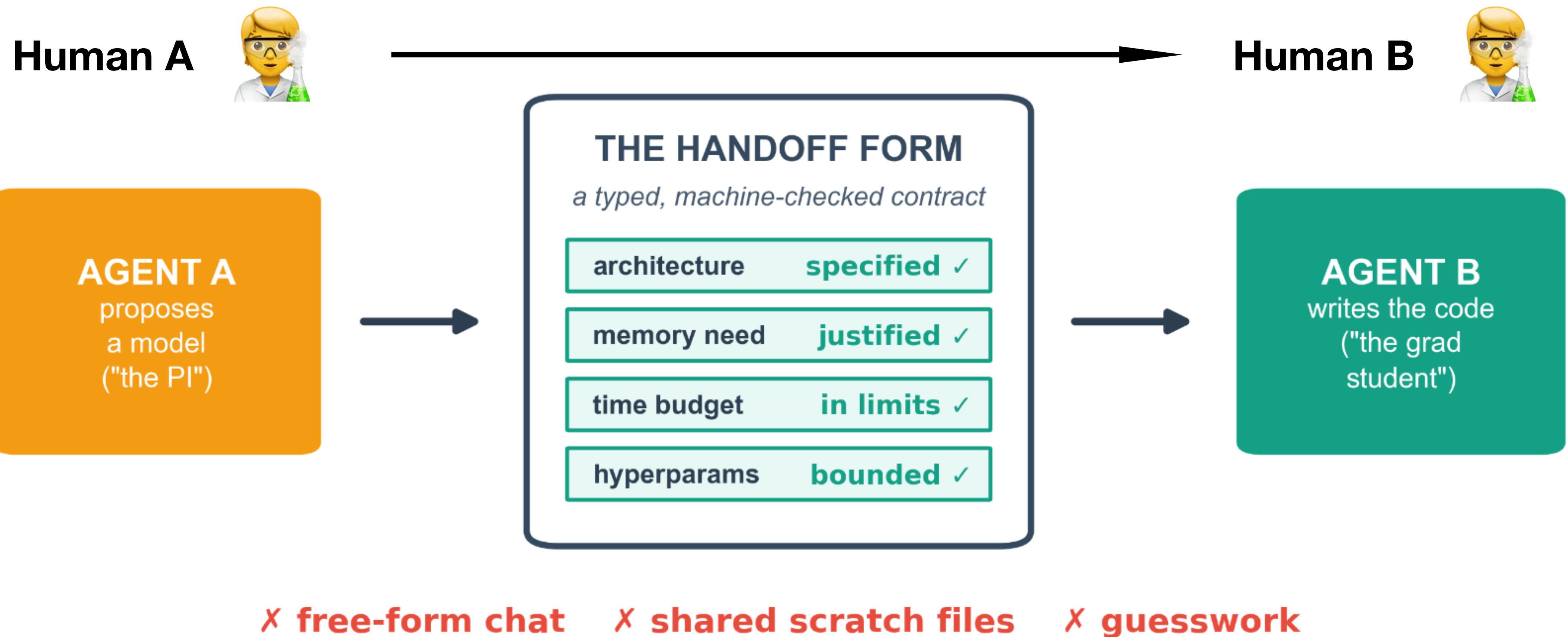
$$\text{Denoising Score} = \log_{5.27} \Lambda$$

SIDERIUS: The Full Structure



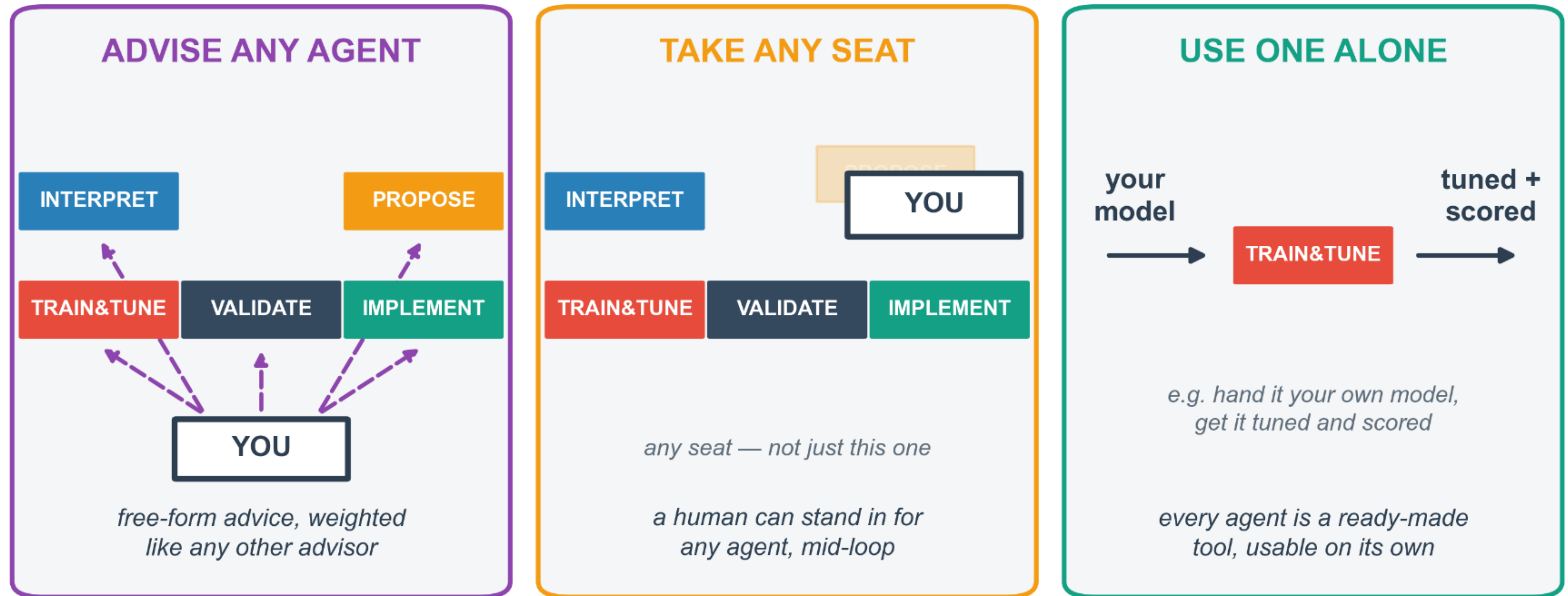
SIDERIUS: The Design Principle

Contract-based, inspired by functional programming



SIDERIUS: Human-in-the-loop

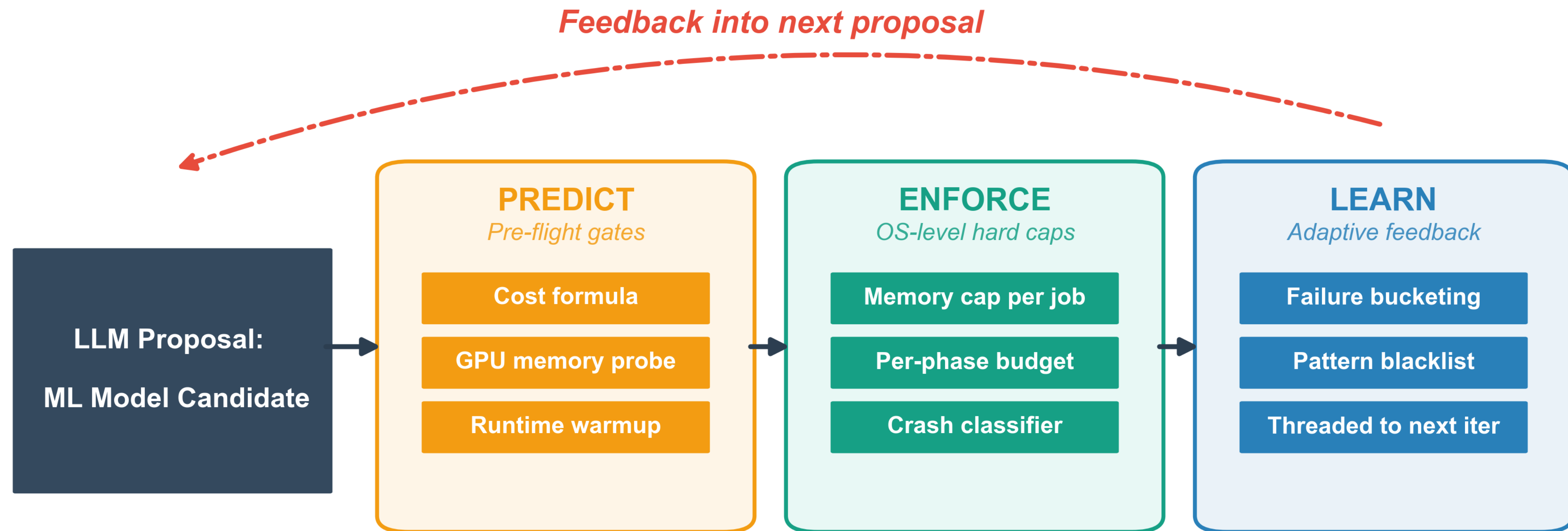
Agents are tools, use them in anyway you like



One typed contract for every seat — humans and agents are interchangeable.

SIDERIUS: Resource-awareness

Multi-layer of Resource-awareness



Every failure is caught, classified, and memorized.

What We Found: Evaluation Matters

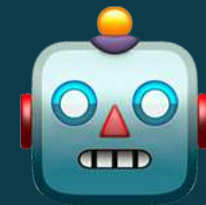
Real-world science is much beyond one single metric

Denoising Score: Signal-to-noise Ratio in the frequency spectrum

What We Found: Evaluation Matters

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Denoising Score: Signal-to-noise Ratio in the frequency spectrum

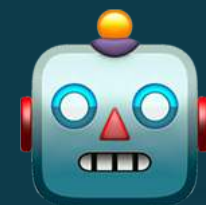


▮ This is easy! Let me eliminate signal and noise altogether, so that they are comparable!

What We Found: Evaluation Matters

Real-world science is much beyond one single metric

Denoising Score: Signal-to-noise Ratio in the frequency spectrum



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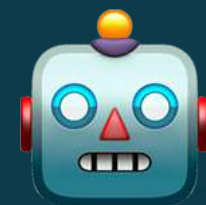
A read hacking mode we found for TIDMAD

- 99.99% of the outputs are 0
- A few non-zero points contribute to the signal frequency

What We Found: Evaluation Matters

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A read hacking mode we found for TIDMAD

- 99.99% of the outputs are 0
- A few non-zero points contribute to the signal frequency

A clear mode collapse here, more stealthy hackings anywhere

Evaluation Matters

A science task come as a full-suite declaration, not a single goal

DECLARED BY THE TASK

TASK PACKAGE

SCIENTIFIC TASK DECLARATION

- data & deliverable
- metrics and their roles
- validity checks
- scientific context

meaning stays outside framework code

WHAT EACH QUANTITY DECIDES



TRAINING OBJECTIVE

what the model is trained to optimize



VALIDATION BEHAVIOR

how learning is going, epoch by epoch



SCIENTIFIC SELECTION

selects the final result

↑ **declared direction**



SECONDARY EVIDENCE

shown to the agents as evidence

never selects

AND WHETHER IT COUNTS AT ALL



VALIDITY

a separate question from quality

GATE 1 · BEFORE SCORING

Is this output scoreable at all?

fails → a recorded refusal

GATE 2 · BETWEEN ROUNDS

Is more compute justified here?

fails → the run is routed away

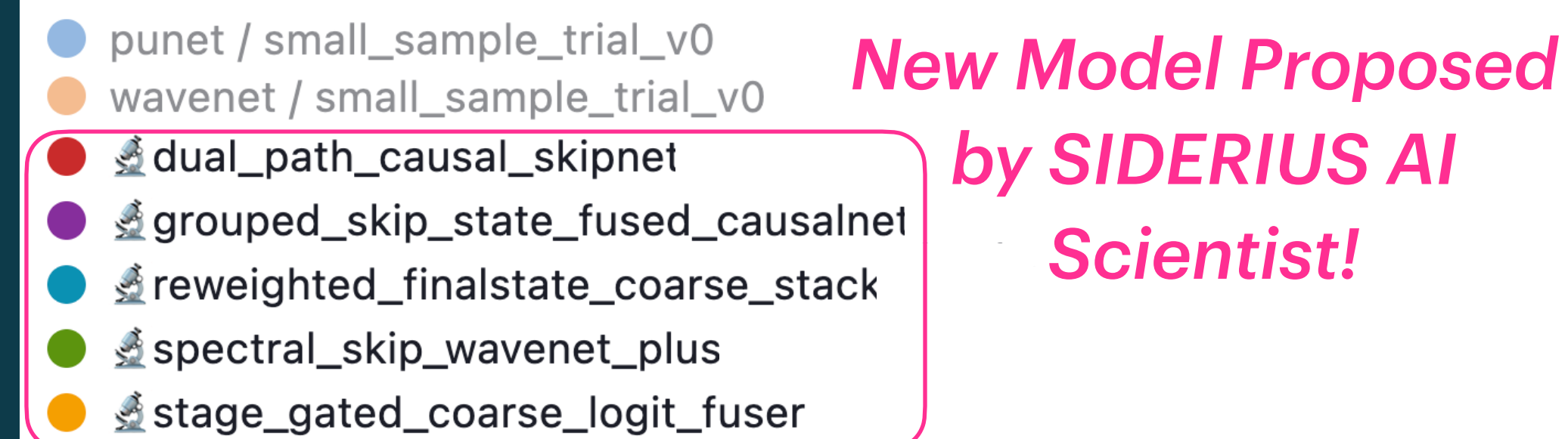
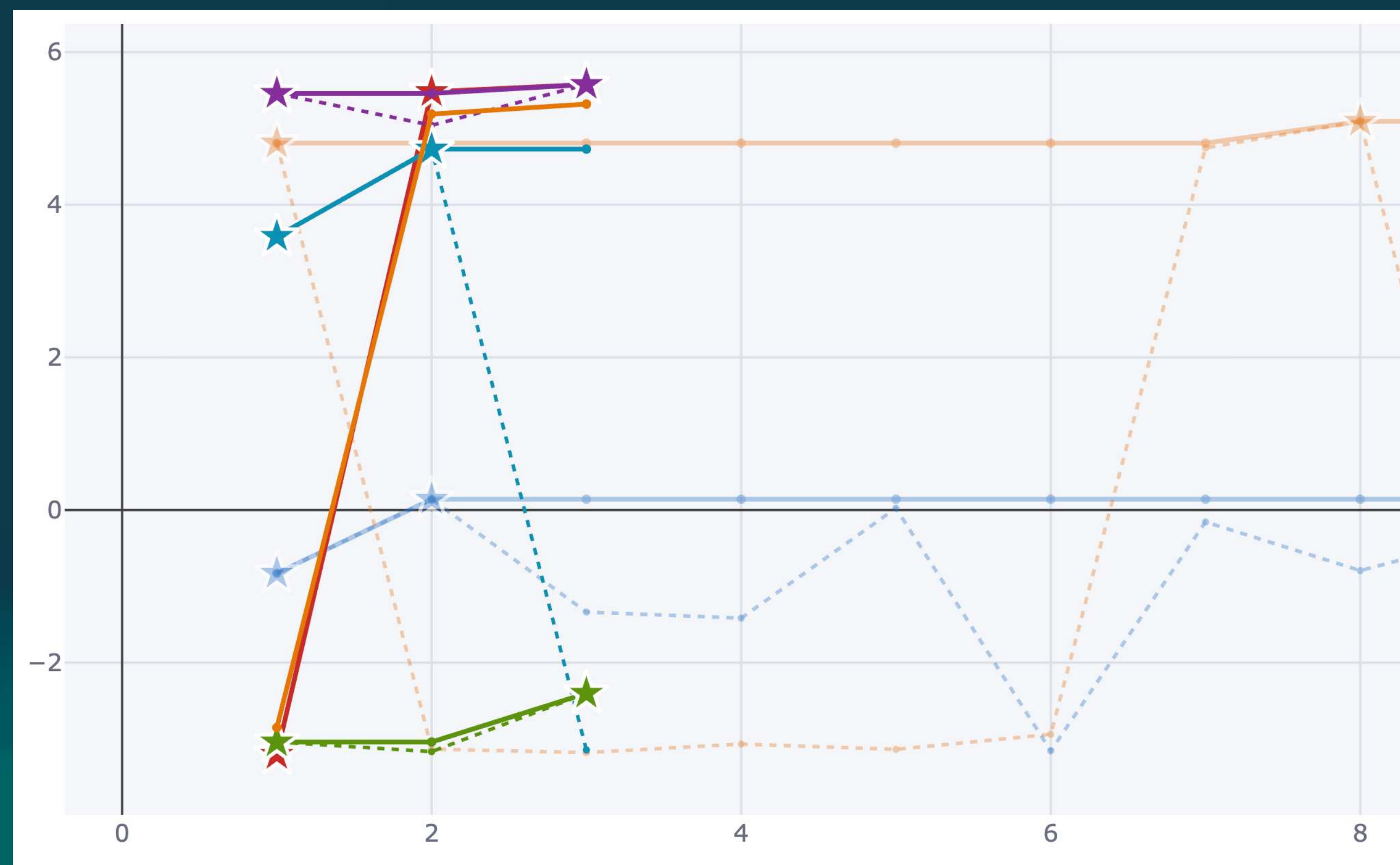
validity

No single number plays all of these roles.

SIDERIUS: Current Progress

Case study: Denoising the Time Series in TIDMAD

Denoising Score



Working example*:

- **Efficient new-ML model proposing, training and tuning**
- **Beat previous paper baselines within a few explorations**

Hyper Parameter Tuning Loop

*on a single TIDMAD file out of 20

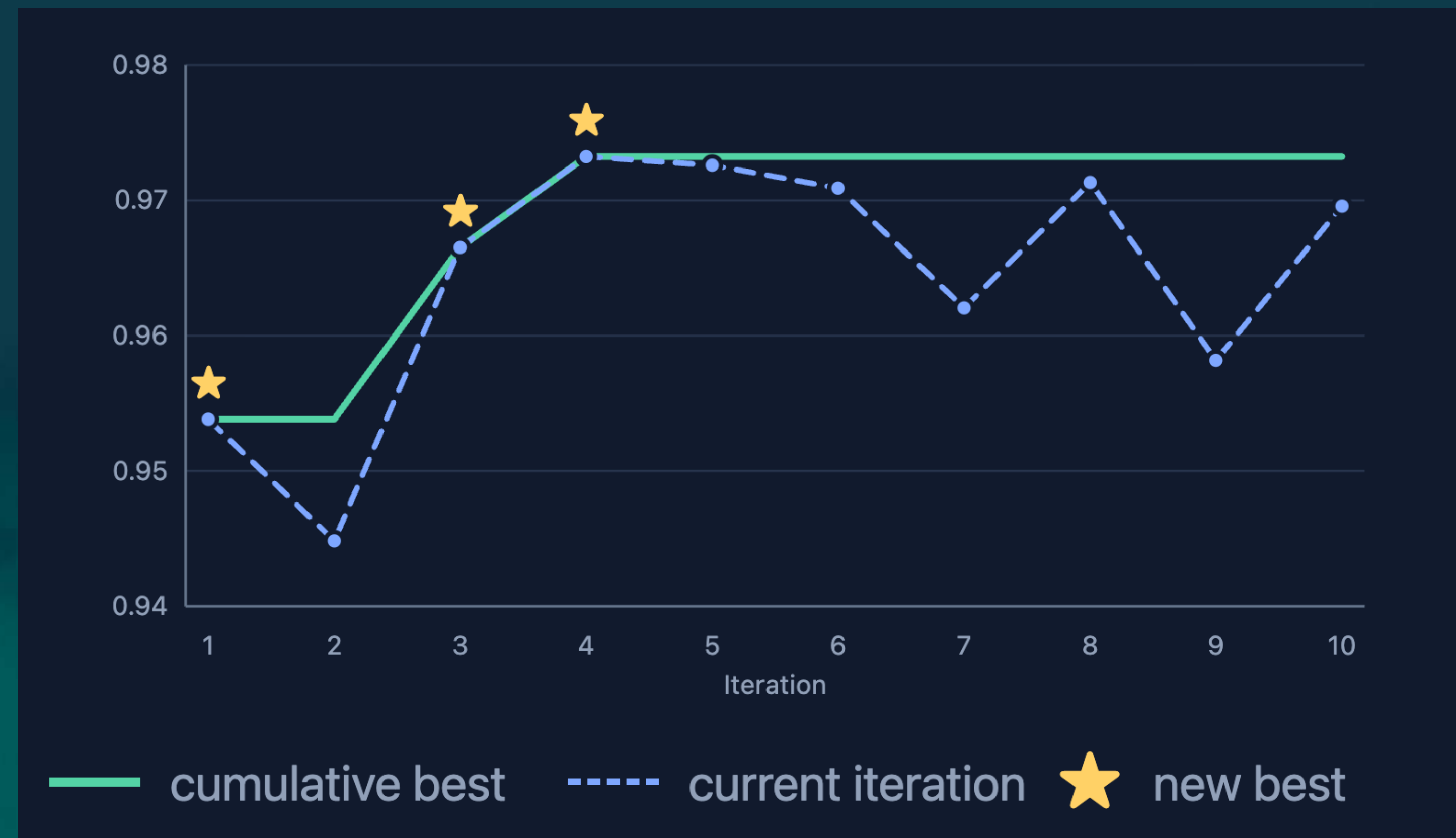
SIDERIUS: Current Progress

Case study: Signal/Background Classification in MAJORANA

Signal: **Single Site**
Background: **Multiple Site**

**Optimizing for:
classification AUC**

- **1 x 5090**
- **One command**
- **within 3 hours**



SIDERIUS: Current Progress

Case study: Signal/Background Classification in SuperNEMO

Signal:

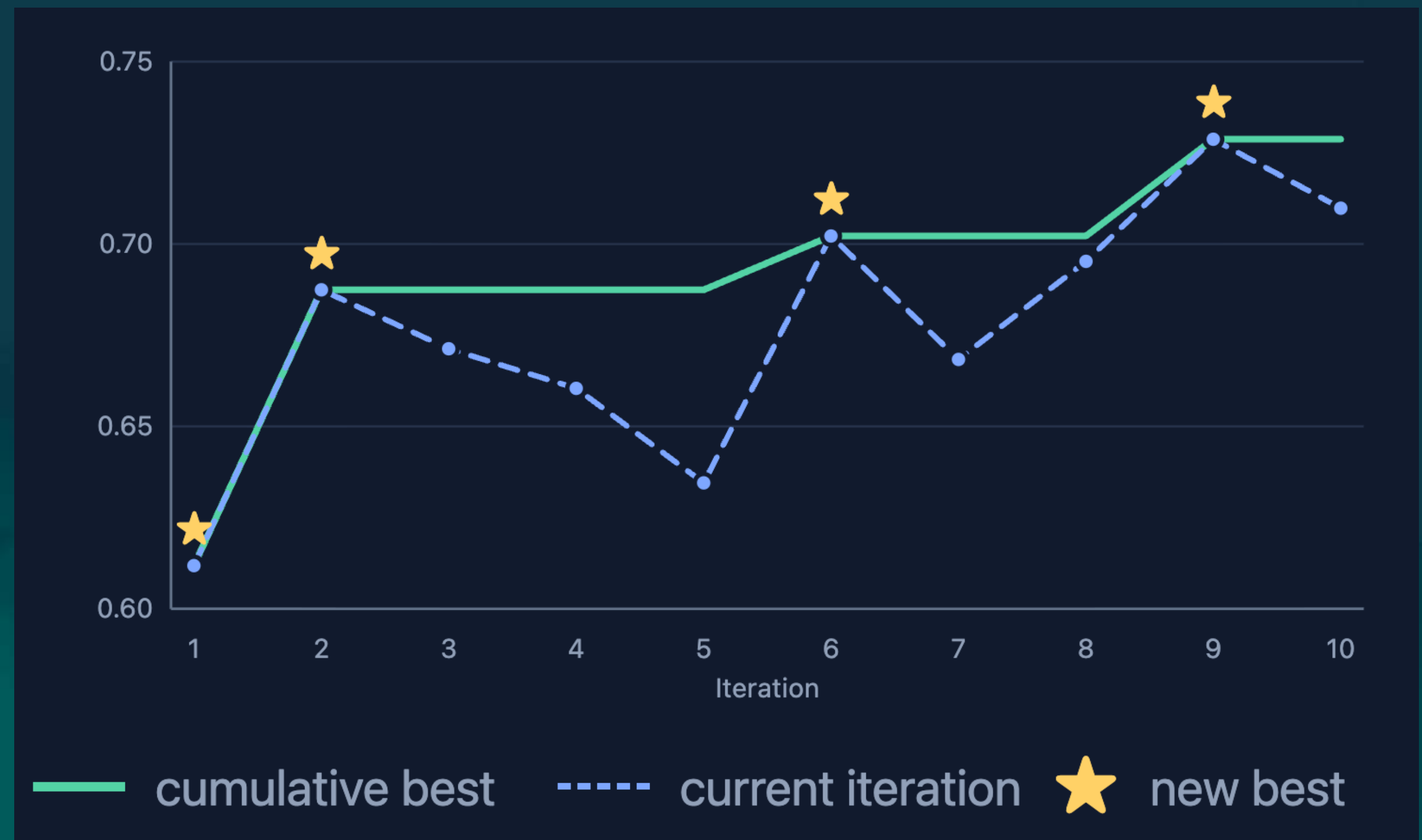
$0\nu\beta\beta$

Background:

$2\nu\beta\beta, \beta + [\gamma \rightarrow \text{Compton}], \text{ etc}$

Optimizing for:
classification AUC

- 1 x 5090
- One command
- within 3 hours



SIDERIUS: Prospects

Ongoing Genesis Mission Program Tasks

	AI TASK	DATA	MAIN CHALLENGE
TIDMAD search for axion dark matter	Denoise the series to recover the signal	SQUID magnetometer 0.8 TB per 24 hours	Noise is complex and varies in time
CUPID TES search for 0vbb decay	Denoise the pulses to sharpen the timing	Heat pulses in a cryogenic crystal	Pile-up background fakes the signal
QEC improve quantum computing	Denoise transient signals in real time	Error-check readouts from the qubits	Must finish within about 1 millisecond

SIDERIUS: What to use for/How to use

Call for internal users and collaborators

- Available data format:
 - Time Series/Image/Video/Point Cloud/etc
- Task type:
 - Classification/Regression/etc

All verified
by concrete
examples ✓

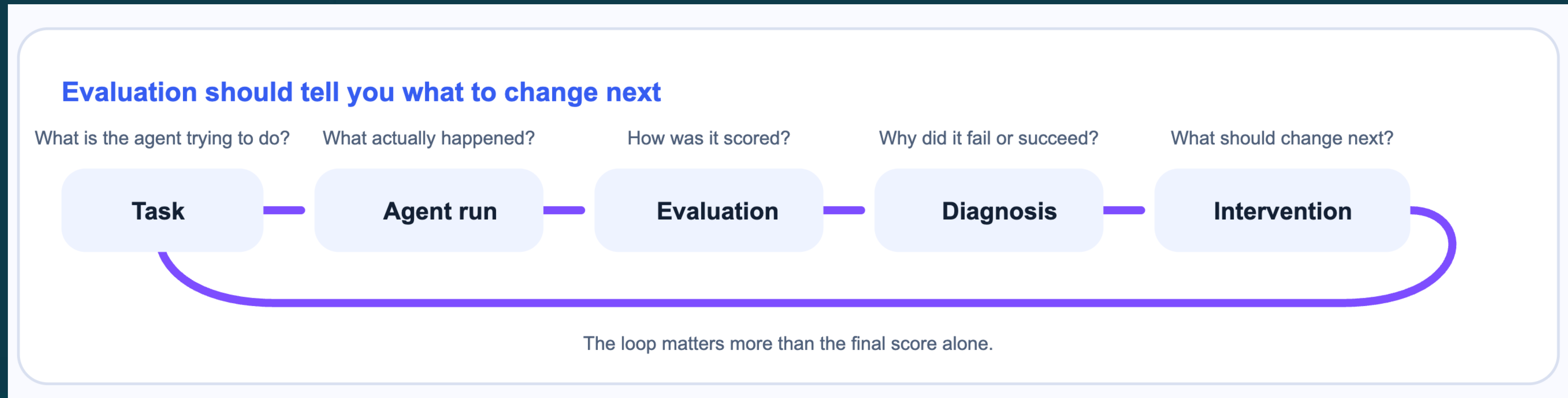
Minimal Example: Hey Claude/Codex, beat SOTA on this dataset, then write a paper for me. + Use the **SIDERIUS** infra to do that!

Now - Oct: Open for internal usage. **Contact us!**

Nov: release **SIDERIUS** as an open-source framework

Echo: Evaluation Matters

Not only for the task, but also for the **agent system itself**



Drifting/Gaming: Evaluate the **failure patterns**, not just the success

Wasting: Evaluate the **efficiency**, not just the outcome

Echo: Evaluation Matters

Join the Effort! All Scientists are Welcome

SciTra

Evaluate Agent **Efficiency** in Cost-sensitive Science/Engineer Task

Build a community for
Scientists-AI Co-evolution

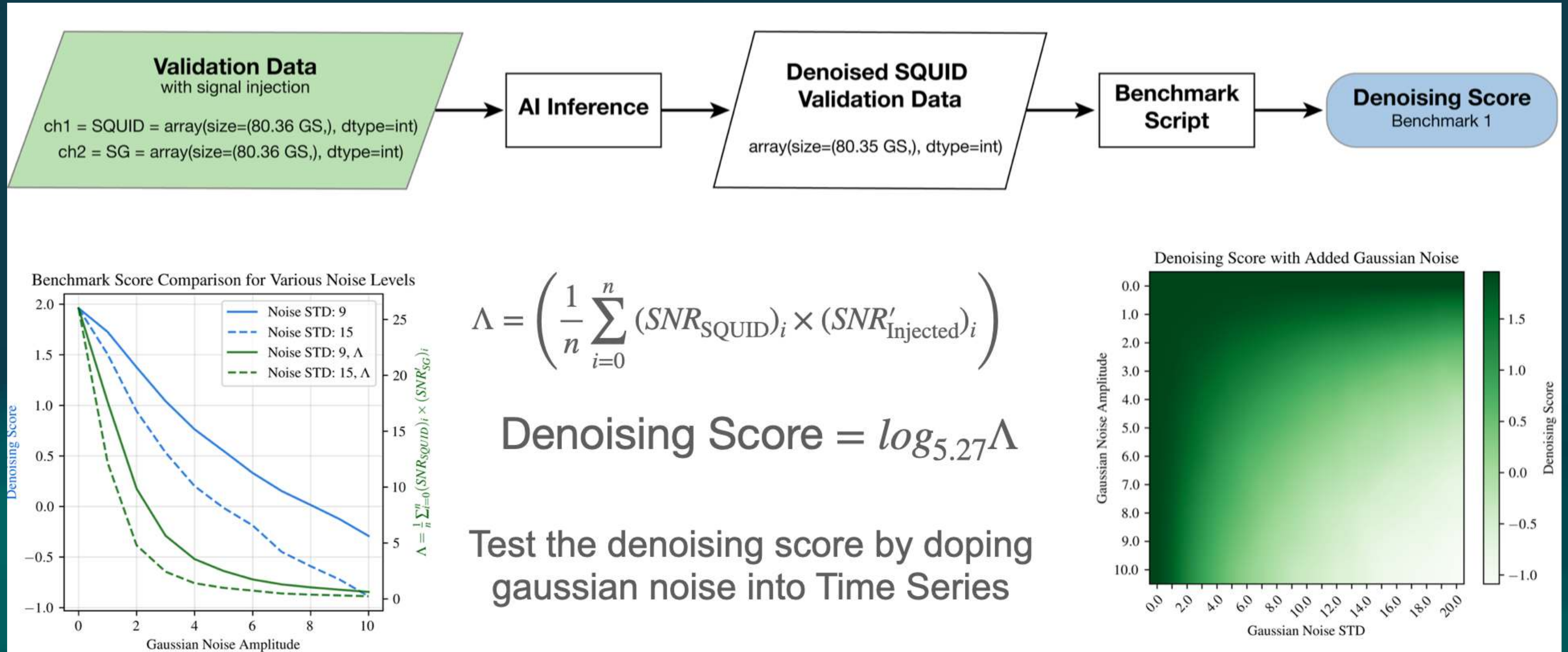


scitra.org

Backups

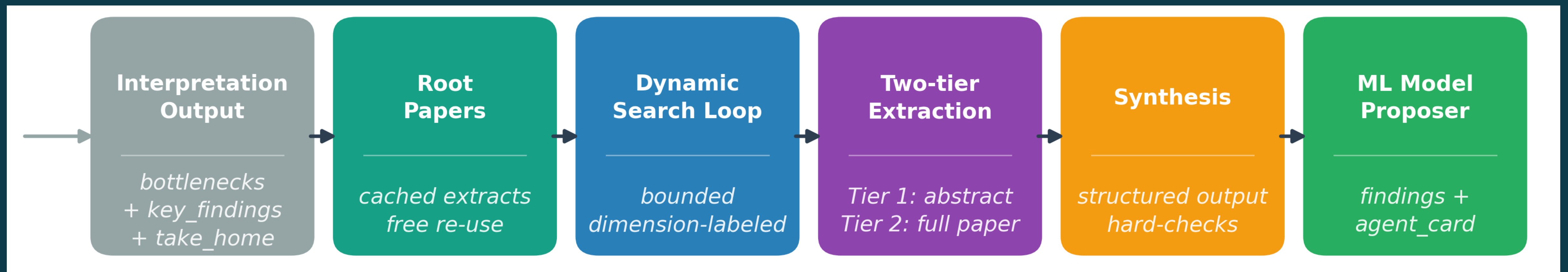
The Background of SIDERIUS

The Metric to Optimize in TIDMAD



SIDERIUS: Literature Review

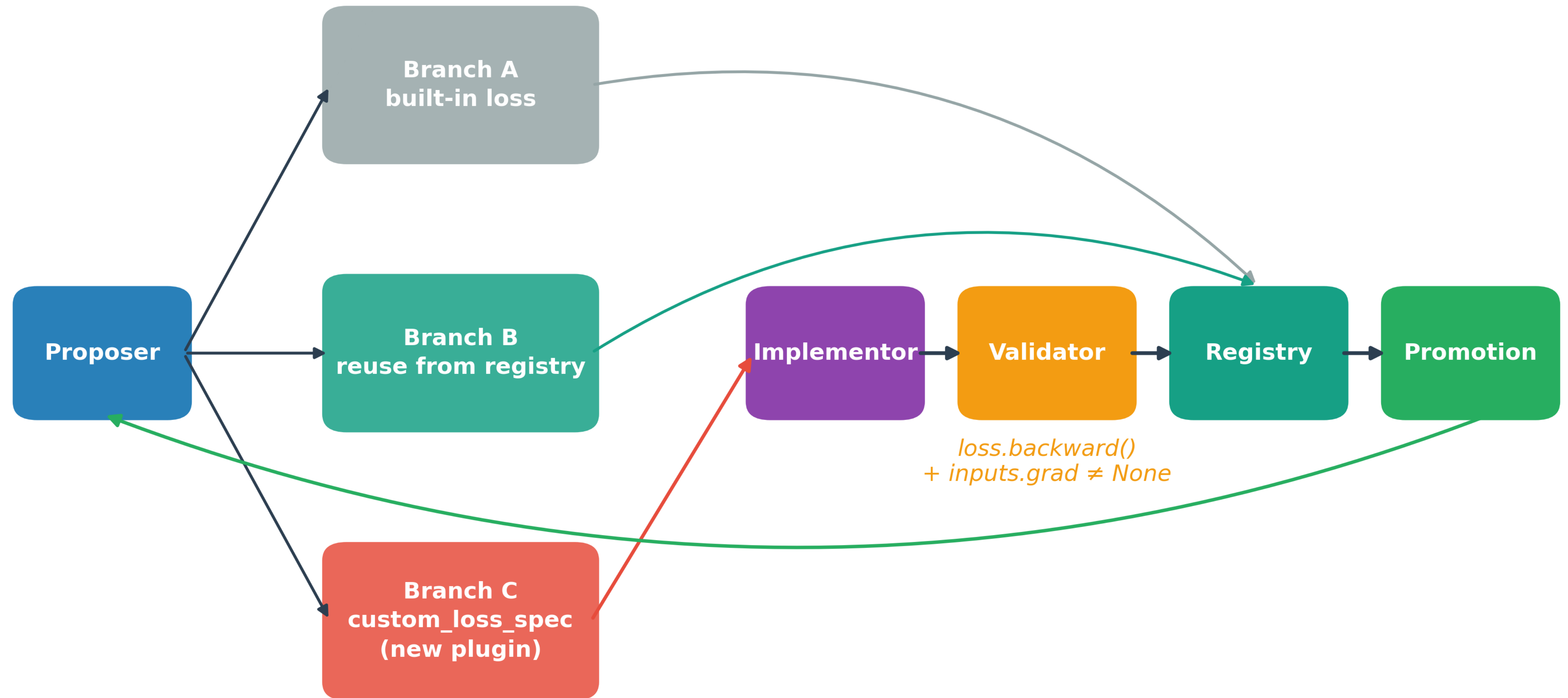
Searching is non-trivial



- **2-Layer Source:** Root paper + Dynamic search
- Search: max round + detention labeled (different aspects)
- **2-Tier Extraction:** Dynamic escalation based on correlation to task
- Protocol to proposer: Synthesis -> finding with confidence + agent_card

SIDERIUS: Toolkit Accumulation

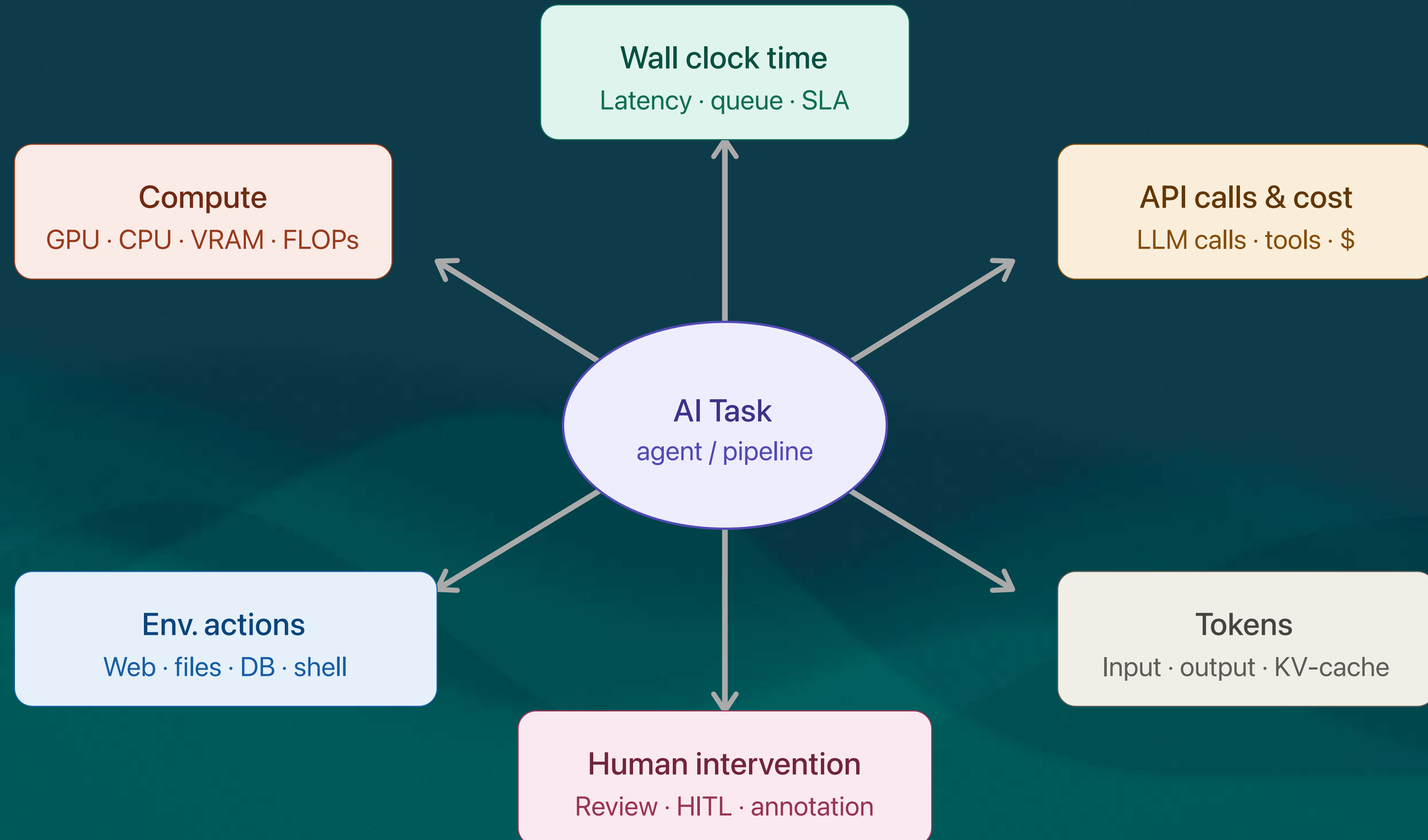
Loss as an example



SciTra: Cost-Aware Science Trajectory

The Bottleneck of Scaling-up: Resource

- End-to-end research workflow is easy to build
- But scientific discovery is fundamentally resource-constrained



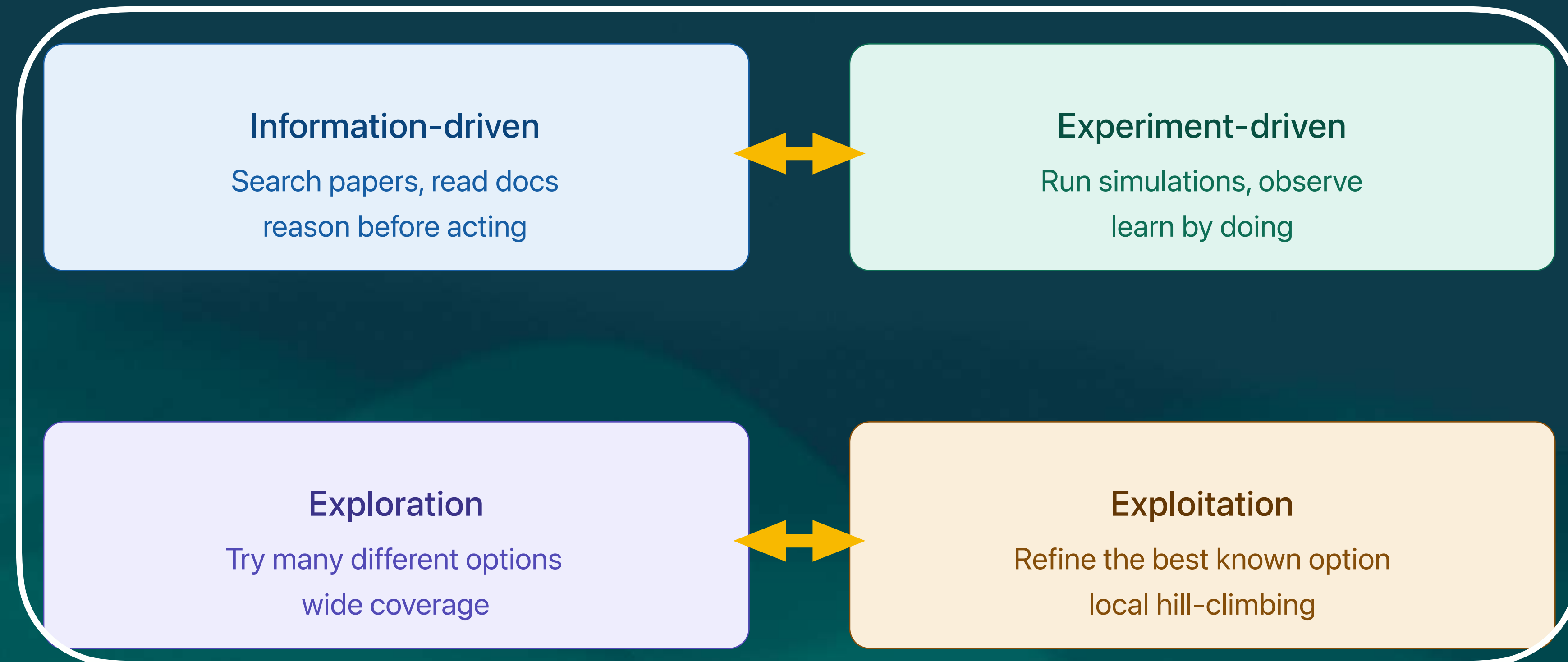
SciTra: Cost-Aware Science Trajectory

From pass@k, to the efficiency evaluation

- A minimal example:

👤: Design a car for me,
make sure the air
resistance is below X at
speed of Y

Strategy Preferences

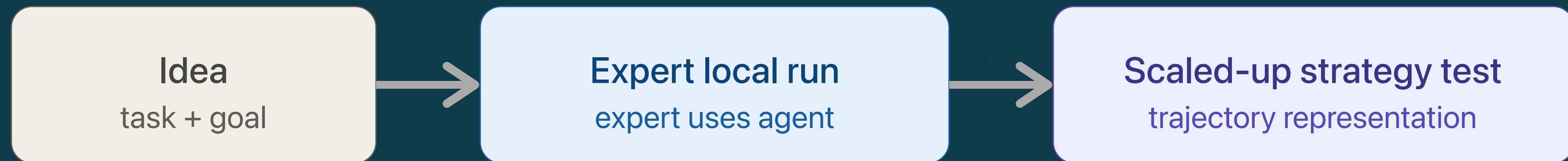


Which is better in terms of efficiency?

Honest answer: agent doesn't know

SciTra: Cost-Aware Science Trajectory

Whole lifecycle of task collection and evaluation



Trajectory and strategy representation: granularity matters

Coarse:



expand

Fine:

